

SHRESHTH VERMA

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PROFILE SUMMARY:

Seeking high-impact Software Engineering / Machine Learning Internship roles in top technology companies. Computer Science undergraduate with strong foundations in Data Structures & Algorithms and hands-on experience in Machine Learning, Deep Learning. Built end-to-end ML systems including regression models, and semantic segmentation architectures.

EDUCATION

Bachelor of Technology (B.Tech) – Computer Science and Engineering
Thapar Institute of Engineering and Technology

Relevant Coursework: Data Structures & Algorithms, Machine Learning, Artificial Intelligence, Operating Systems, Database Management System.

TECHNICAL SKILLS

- **Programming** : Python, C, C++, SQL
- **Machine Learning** : Regression, Classification, ANN, CNN, U-Net, Hyperparameter Tuning
- **MLOps & Tools** : MLflow, DVC, Git, Experiment Tracking, Model Versioning
- **Libraries** : PyTorch, Scikit-learn, NumPy, Pandas, Matplotlib
- **Core CS** : Data Structures & Algorithms, Object Oriented Programming, Feature Engineering, Model Evaluation Metrics

KEY PROJECTS

- **Air Quality Index (AQI) Prediction System:** Built regression models with advanced feature engineering and hyperparameter optimization. Evaluated performance using RMSE and R^2 metrics.
- **Customer Churn Prediction using ANN:** Developed neural network classifier with preprocessing, feature scaling, and performance evaluation using Precision, Recall, and F1-score.
- **Image Segmentation using U-Net (PyTorch):** Implemented U-Net and Nested U-Net architectures with data augmentation and optimized loss functions for semantic segmentation.
- **MLOps Workflow Implementation:** Integrated MLflow and DVC for experiment tracking, dataset versioning, and reproducible ML pipelines.
- **Malaria Cell Classification – Architecture Comparison:** Compared 5 deep learning architectures on ~27,000 malaria cell images using PyTorch. Custom CNN achieved 95.5% accuracy and ROC-AUC of 0.991, outperforming pretrained models.

ACHIEVEMENTS & LEADERSHIP

- Participant – Smart India Hackathon (collaborated in high-pressure, team-based problem-solving environment).
- Actively strengthening DSA and problem-solving skills for top-tier SDE/ML internship roles